Use the force diagram to find acceleration

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Component / ES: componente / PT: componente

The downhill part of weight. A way to calculate — not an extra force.

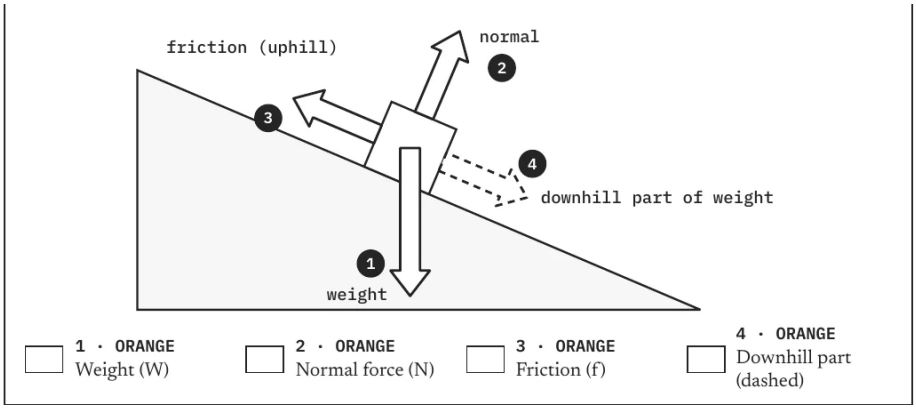
Also on your note page: Normal force, Weight, Acceleration direction.

B READ THE RELATIONSHIP

A component is the part of one vector along one chosen axis.

Resolve one force along two perpendicular axes

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A 2 kg block has a 7 N downhill weight component and 3 N of friction uphill. Find its acceleration.

Given m = 2 kg 7 N downhill 3 N uphill

Rule a = F net ÷ m

Work F net = 7 − 3 = 4 N downhill a = 4 ÷ 2

Answer 2 m/s² downhill

E SAVE FOR THE FINAL MEETING

From the diagram, the net force gives XJ's acceleration and its direction. Get the net force's direction right and you know which way its path will bend.

Read the physics: Component

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

On a ramp, gravity still points straight down. The normal force points straight out of the surface. Choose one axis along the ramp and one perpendicular to it.

02 HOW IT WORKS

Split weight into two parts. The parallel part pulls downhill; the perpendicular part presses into the ramp. Calculate with the parts or with the original weight, never with both together.

03 WHAT TO CHECK

Subtract uphill friction from the downhill force component. If the block stays on the ramp, its acceleration perpendicular to the ramp is zero.

04 TRY THE EXAMPLE FROM YOUR NOTES

A 2 kg block has a 7 N downhill weight component and 3 N of friction uphill. Find its acceleration. 2 m/s² downhill.

05 CONNECT TO XJ

From the diagram, the net force gives XJ's acceleration and its direction. Get the net force's direction right and you know which way its path will bend.

SAY IT BEFORE YOU WRITE

Why would adding both weight and its components be a mistake? Along the ramp I use ___. Across the ramp I use ___.

Build the word. Draw the idea.

STUDENT CHOICE / FRONT / OPENING MEETING Name _____

A FRAYER MODEL

Component / ES: componente / PT: componente

Say it in any language. Draw or use a few words. Use the notes to check your idea.

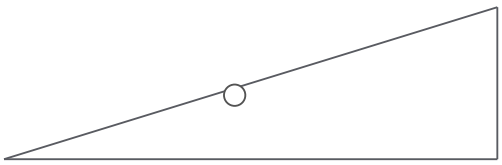
Meaning	How can you tell?
To me, component means …	Which part of the weight pulls the block down the ramp? I use the part of the weight that points ___.
Example or sketch	**Non-example**
I see it when …	This looks close, but …

B DRAW WITH YOUR TEACHER

Draw the real weight once, then draw its two components in a separate box. Circle the component along the ramp.

Labels you can use: parallel · perpendicular · downhill.

Real forces



Components

Tell a partner: Along the ramp I use ___. Across the ramp I use ___.

On assignment: Component

SECTION E / BACK / FINAL MEETING ONLY Name _____

A navigation graphic lists a gravitational force AND its two components as three separate pulls on XJ.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Navigation engineer**

Build an overlay using the component view. Indicate the full arrow it replaces before releasing the values to the calculator.

☐ **Map graphics editor**

Sketch two camera views: the real pull, then its components. Give viewers a caption that prevents counting the same pull twice.

Word bank: parallel · perpendicular · downhill.

Try: “Our desk recommends ___. This mark shows ___.”

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

“We first said ___. Now we ___ because ___.”